

SOFTWARE REQUIREMENTS SPECIFICATION

AutoQP

AI Based Question Paper Generation using $\text{\LaTeX}$

Prepared in accordance with IEEE Std 830-1998

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1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements of AutoQP, a centralized web platform that allows instructors to generate, extract, review, store, and assemble exam questions, and compile them into publication-quality PDF question papers via an automated L^AT_EX-based document engine. This document is intended to serve as the authoritative reference for what the system must do, for use by the development team, the course instructor acting as project sponsor, and any future maintainer of the codebase. It specifies required behavior, not implementation: database schema design, API contracts, and coding conventions are addressed by separate project artifacts and are outside the scope of this document, except where a specific technology choice is itself a binding constraint on requirements (see Section 2.5).

1.2 Document Conventions

Requirements are numbered REQ-**<feature>**.**<index>** (e.g. REQ-1.2) and grouped by system feature in Section 4. Each feature is assigned one of three priority levels, mapped from the project’s MoSCoW prioritization:

Priority in this SRS	MoSCoW equivalent
High	Must-Have
Medium	Should-Have
Low	Could-Have

Features marked *Wish List* in the originating project outline are explicitly excluded from this SRS (see Section 1.4) and are not assigned requirement IDs. The keywords **shall** (mandatory), **should** (recommended but not mandatory), and **may** (optional) are used per IEEE 830 convention to indicate the binding strength of each requirement.

1.3 Product Scope

AutoQP replaces manual, word-processor-based exam typesetting with a centralized, subject-partitioned question bank and an automated document compilation pipeline. In scope: AI-assisted question generation, extraction of questions from pasted text and screenshots, a human-review staging area, a searchable shared repository, automated exam blueprint assembly, and LaTeX-based PDF compilation with a raw markup source. Out of scope for this version: any student-facing functionality (submission, auto-grading, a student portal), multi-institution moderation/approval chains, full-page computer-vision ingestion of scanned multi-column exams, and semantic duplicate detection via embeddings. These exclusions correspond to the “Wish List” tier of the originating project outline and were confirmed out of scope by the project’s feasibility review (see Section 1.5).

1.4 References

- IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*.
- Project Outline and MoSCoW Feature Breakdown (originating requirements brief supplied by the project sponsor).
- Project Feasibility Report (Technical, Operational, Economic, Time, Social, Legal, Market, and Resource feasibility findings).

2 Overall Description

2.1 Product Perspective

AutoQP is a new, standalone product; it is not a modification of an existing system. It is intended to replace an informal, ad hoc workflow (instructors typesetting exams individually in Microsoft Word or similar word processors) with a shared, centrally managed platform. Figure 1 shows the system’s high-level architecture: a browser-based client communicates exclusively with an application server, which is the sole component authorized to communicate with the database, the AI provider, and the document compilation service.

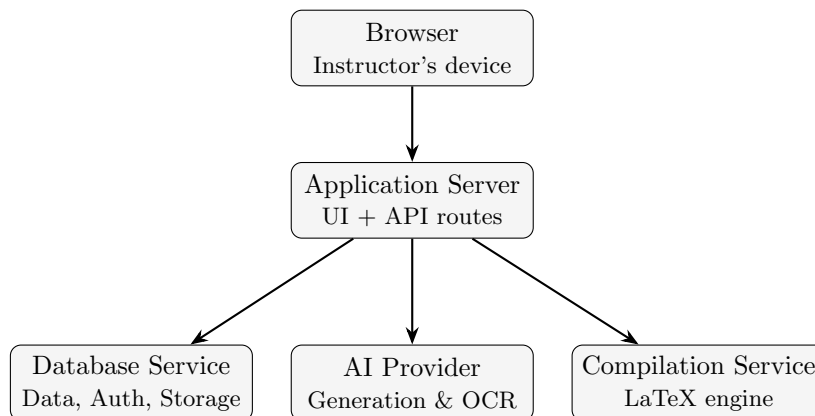


Figure 1: High-level system architecture.

2.2 Product Functions

At a high level, the system shall allow an authenticated instructor to: generate or extract candidate questions; review and edit those candidates before they enter a shared bank; search and filter the bank by subject, topic, difficulty, and marks; assemble a paper automatically from a mark pattern; and compile the assembled paper into a downloadable PDF and its raw LaTeX source. A full breakdown appears in Section 4.

2.3 User Classes and Characteristics

- | | |
|----------------------|--|
| Instructor | The primary user class. Assumed to be subject-matter experts with no assumed programming or LaTeX background. Interacts with the system to generate, review, bank, and assemble questions into papers scoped to the subjects they are assigned to. |
| Administrator | A secondary user class responsible for creating subjects and assigning instructors to them. Assumed to have the same technical background as an Instructor; distinguished only by elevated permissions, not by technical skill. |

2.4 Operating Environment

The client application shall run in current versions of major evergreen web browsers (Chrome, Firefox, Edge, Safari) with no browser plugin requirements. The application server shall run on a standard managed Node.js hosting platform. The database shall be a hosted relational database service. The compilation service shall run in a containerized Linux environment with no outbound network access during the compilation step itself (see REQ-7.4, Section 5.3).

2.5 Design and Implementation Constraints

- The system shall use a relational database as its primary data store, with subject-scoped access control enforced at the database layer, not solely in application code.
- The system shall use a headless, sandboxable LaTeX engine together with the `exam` LaTeX document class for PDF compilation; shell-escape shall be disabled in all compilation contexts.
- Per the project's Time Feasibility finding, the first delivery milestone shall implement High-priority requirements only; Medium- and Low-priority requirements are documented for completeness but are not committed to the first milestone.
- The development team consists of four members completing this project within a single academic semester; this constrains the achievable scope and is the basis for the priority-based phasing described above.

2.6 User Documentation

No end-user documentation exists at the time of writing. An instructor onboarding guide is recommended as future work once the High-priority feature set is stable, but is not itself a requirement of this specification.

2.7 Assumptions and Dependencies

- The system depends on the continued availability and reasonable response latency of a third-party multimodal AI provider for question generation and screenshot/diagram extraction.
- The system assumes instructors are willing to review AI-generated or AI-extracted content before it is committed to the shared bank; no requirement in this document permits AI output to bypass human review.
- Development is assumed to occur within the free-tier usage limits of the chosen hosting, database, and compilation platforms; the project's Economic Feasibility finding requires a usage-based cost model before any deployment beyond the course submission.

3 External Interface Requirements

3.1 User Interfaces

The system shall provide a web-based graphical interface requiring no installed software beyond a standard browser. Instructors without LaTeX experience shall be able to complete the full workflow — generation through compiled PDF — without directly editing raw LaTeX markup; direct markup editing shall be available only as an optional, clearly-labelled advanced path.

3.2 Hardware Interfaces

None. The system has no direct hardware interface requirements beyond standard consumer computing devices capable of running a modern web browser.

3.3 Software Interfaces

- **Database service:** the system's primary data store, authentication provider, and object storage for uploaded screenshots and compiled output files.
- **AI provider:** used for AI-assisted question generation and for multimodal extraction of text, mathematical notation, and diagrams from screenshots.

- **LaTeX compilation engine:** invoked by the application server to compile assembled papers into PDF output.

3.4 Communications Interfaces

All communication between the browser client and the application server shall occur over HTTPS. All communication between the application server and the database service, the AI provider, and the compilation service shall likewise occur over encrypted connections.

4 System Features

This section enumerates system features grouped by MoSCoW-derived priority. High-priority features constitute the committed scope of the first delivery milestone (see Section 2.5).

4.1 Role and Subject Scoping (High)

Priority: High

Description: Instructors authenticate and are restricted to viewing and editing question data only within subjects they are explicitly assigned to.

REQ-1.1 The system shall require authentication before any question bank data is accessible.

REQ-1.2 The system shall associate each instructor account with zero or more subjects via an explicit assignment record.

REQ-1.3 The system shall deny read and write access to question data for any subject an instructor is not assigned to, enforced at the database access-control layer in addition to the application layer.

REQ-1.4 Administrator accounts shall be able to create subjects and assign instructors to them.

4.2 AI-Assisted Question Generation (High)

Priority: High

Description: Instructors supply generation parameters and receive formatted candidate questions with embedded mathematical notation.

REQ-2.1 The system shall accept topic, difficulty, mark weight, and question count as generation parameters.

REQ-2.2 The system shall return generated questions with mathematical notation rendered correctly when previewed.

REQ-2.3 The system shall route every generated question through the Staging Sandbox (Section 4.4) before it may enter the shared bank; no generated question shall be committed automatically.

4.3 Multimodal Extraction (High)

Priority: High

Description: Instructors submit pasted text or a screenshot, which the system parses into either structured text/mathematics or a preserved image asset, depending on content type.

REQ-3.1 The system shall accept pasted unstructured text and parse it into question text, answer options (where applicable), and mathematical markup.

- REQ-3.2** The system shall accept an uploaded screenshot and extract text and mathematical content when the image depicts an equation or textual question.
- REQ-3.3** The system shall preserve an uploaded screenshot as a stored image asset, rather than attempting text extraction, when the image depicts a diagram, graph, or figure not reducible to text.
- REQ-3.4** Image assets associated with a question shall be embedded in the compiled PDF output at the position of that question.

4.4 Staging Sandbox (High)

Priority: High

Description: A temporary, per-subject review area in which AI-generated or AI-extracted questions are edited, accepted into the bank, or discarded.

- REQ-4.1** The system shall allow an instructor to edit any field of a staged question before committing it to the bank.
- REQ-4.2** The system shall permit a staged question to be incomplete (e.g. missing a topic tag) while under review.
- REQ-4.3** The system shall remove a staged question from the staging area upon explicit discard or upon successful commit to the bank; staged questions shall not persist indefinitely.
- REQ-4.4** The system shall enforce all bank-level field requirements (Section 4.5) at the point of commit, rejecting a commit that does not satisfy them.

4.5 Subject Question Repository (High)

Priority: High

Description: Persistent, searchable storage of committed questions categorized by subject, topic, difficulty, and marks.

- REQ-5.1** The system shall require every committed question to have a subject, a question type, a difficulty level, a mark value greater than zero, and non-empty question text.
- REQ-5.2** The system shall allow filtering the repository by subject, topic, difficulty, and mark value, individually or in combination.
- REQ-5.3** The system shall support free-text keyword search over question text.
- REQ-5.4** The system shall support at minimum three question types — multiple choice, numerical, and long-answer — and shall accommodate the addition of further question types without requiring a database schema migration.

4.6 Pattern-Based Blueprint Generator (High)

Priority: High

Description: Automated, non-repeating question selection against an instructor-specified mark pattern.

- REQ-6.1** The system shall accept a mark pattern expressed as a set of (question count, mark value) pairs.
- REQ-6.2** The system shall select questions satisfying the pattern without selecting the same question more than once within a single generated paper.
- REQ-6.3** The system shall only select questions from the committed repository (Section 4.5); staged, uncommitted questions shall never be selectable.

4.7 Automated Document Compilation (High)

Priority: High

Description: Compilation of an assembled paper’s metadata and questions into a downloadable PDF and its raw LaTeX source.

REQ-7.1 The system shall inject paper metadata (title, time limit, instructions) and all assembled questions into a document template prior to compilation.

REQ-7.2 The system shall produce a downloadable PDF file and a downloadable raw LaTeX source archive upon successful compilation.

REQ-7.3 The system shall report a compilation failure to the instructor without a partial or corrupted output file being made available for download.

REQ-7.4 The compilation process shall execute with shell-escape disabled and without out-bound network access at the point of invoking the LaTeX engine.

4.8 Two-Panel Visual Paper Assembler (Medium)

Priority: Medium

Description: An interactive canvas for manually composing a paper by dragging questions from a searchable repository panel into an organized, sectioned exam canvas.

REQ-8.1 The system should allow a question to be moved from the repository panel into a named section of the exam canvas via drag-and-drop.

REQ-8.2 The system should allow reordering of questions within a section.

REQ-8.3 The system should display a running total of marks per section and for the paper as a whole, updated as questions are added, removed, or reordered.

4.9 Saved Papers History (Medium)

Priority: Medium

Description: Preservation of a compiled paper’s exact question content at the time of assembly, independent of later edits to the source questions in the bank.

REQ-9.1 The system should store a frozen copy of each question’s content at the time it is added to a paper, rather than a reference re-read from the live bank at render time.

REQ-9.2 Editing a bank question after it has been used in a paper should not alter that paper’s previously compiled or assembled content.

REQ-9.3 Deleting a bank question should not delete or corrupt any paper that previously used it.

4.10 Concurrency and Modification Auditing (Medium)

Priority: Medium

Description: Visibility into who last modified a shared question, protection against silently overwriting concurrent edits, and the ability to bring a paper’s frozen question content up to date with the bank on request.

REQ-10.1 The system should display the identity of the last editor and the last-modified timestamp on each bank question.

REQ-10.2 The system should reject an update to a bank question if the record has been modified since the editor last loaded it, and should inform the editor of the conflict rather than silently overwriting the concurrent change.

REQ-10.3 The system should indicate, per question within an assembled paper, whether the bank's current version of that question differs from the version frozen in the paper.

REQ-10.4 The system should allow an instructor to explicitly refresh a paper's frozen question content to match the bank's current version, on a per-question basis, as an instructor-initiated action rather than an automatic background process. This action shall not be available for a paper question whose original bank source has been deleted.

4.11 Question Forking / Duplication (Medium)

Priority: Medium

Description: The ability to clone a shared bank question into a private, editable copy for personal tweaks, without altering the shared original. Reclassified from Low to Medium priority in Version 1.1 of this document: without this capability, an instructor wanting a personalized variant of a shared question has an incentive to edit the shared question in place instead, which risks the same kind of unintended shared-state corruption that Concurrency Auditing (Section 4.10) exists to prevent.

REQ-11.1 The system should allow an instructor to create an editable private copy of any bank question they can view, without modifying the original question.

REQ-11.2 A forked question should retain a reference to the bank question it was cloned from, for lineage and audit purposes, without that reference constraining independent edits to the fork.

4.12 In-Browser Markup Editor (Medium)

Priority: Medium

Description: An optional code-level editor for advanced users to fine-tune raw document markup before compilation.

REQ-12.1 The system should provide an optional view in which the raw LaTeX source of an assembled paper can be edited directly prior to compilation.

REQ-12.2 Manual edits made in this view should be preserved through a subsequent compilation attempt unless the instructor explicitly regenerates the source from the paper's structured content.

4.13 Live Document Preview (Medium)

Priority: Medium

Description: An embedded, in-browser PDF preview that updates upon compilation without a manual download.

REQ-13.1 The system should render a compiled PDF inline in the browser without requiring the instructor to download the file first.

4.14 Document Parsing from .docx / PDF (Low)

Priority: Low

REQ-14.1 The system may support bulk extraction of candidate questions from uploaded .docx or native (non-scanned) PDF files, routed through the Staging Sandbox as with any other extraction source.

4.15 Multiple Layout Presets (Low)

Priority: Low

REQ-15.1 The system may offer a selection of document layout templates (e.g. midterm, quiz slip, final exam with roll-number grid) selectable at compilation time.

4.16 Automated Marking Guide Generation (Low)

Priority: Low

REQ-16.1 The system may generate a parallel solutions or grading-rubric document alongside a compiled question paper.

5 Other Nonfunctional Requirements

5.1 Performance Requirements

- The system shall return repository search and filter results for a subject with up to 2,000 questions within 2 seconds under normal load.
- The system shall complete compilation of a paper of up to 30 questions and 5 embedded images within 30 seconds under normal load.

5.2 Safety Requirements

Not applicable in the physical-safety sense; this system does not control physical equipment. Data-safety requirements are addressed under Section 5.3 and Section 4.9 (Saved Papers History).

5.3 Security Requirements

- Subject-scoped access control shall be enforced at the database access-control layer, not application-layer checks alone.
- AI-generated or AI-extracted content shall never be committed to the shared question bank without passing through instructor review in the Staging Sandbox.
- The document compilation process shall run with shell-escape disabled and without outbound network access at the point of compilation, to contain the risk of executing untrusted generated content.
- All data in transit between system components shall be encrypted (Section 3.4).

5.4 Software Quality Attributes

- **Usability:** An instructor with no LaTeX background shall be able to complete the generation-to-compiled-PDF workflow without directly editing raw markup.
- **Maintainability:** All code that produces or consumes question data shall conform to a single, consistently applied question data shape across every producer and consumer in the system; any change to that shape shall be propagated to all of them in the same change.
- **Portability:** The client application shall function correctly on the current and immediately preceding major release of each major evergreen browser.

5.5 Business Rules

- Only questions committed to the repository (Section 4.5) may be selected by the Blueprint Generator (Section 4.6); staged, uncommitted questions are never eligible.

- A compiled paper’s recorded question content shall not change as a result of edits to the source bank question, except through the explicit refresh action defined in REQ-10.4.
- An instructor may only view or modify question data for subjects they are explicitly assigned to (REQ-1.3).

Appendix A: Data Model Summary

The following summarizes the core entities and relationships required by this specification, at a level sufficient to reason about the requirements above. Implementation-level detail — exact data types, indexes, and constraints — is a design concern outside the scope of this document.

From	To	Relationship
Instructor	Subject	Many-to-many, via an explicit assignment record
Subject	Topic	One-to-many
Subject	Question	One-to-many
Topic	Question	One-to-many (optional)
Question	Question	Self-referencing, one-to-many (fork lineage)
Subject	Paper	One-to-many
Paper	Paper Section	One-to-many
Paper Section	Paper Question	One-to-many
Question	Paper Question	One-to-many, soft reference only (a frozen snapshot is the source of truth for rendering, not this link)

Appendix B: Glossary

Bank The committed, persistent repository of reviewed questions, as distinct from the Staging Sandbox.

Blueprint A mark-based pattern (e.g. five one-mark questions, five two-mark questions) used to auto-select a paper’s questions.

Snapshot A frozen copy of a question’s content stored against a specific paper, immune to later edits of the source bank question.

Staging Sandbox A temporary, per-subject review area for AI-generated or AI-extracted questions prior to bank commitment.